20/7/26 Experiment - 1 20/7/26

AIM: Study of various Network commands used in Linux and windows.

COMMANDS:

1) arp -a.

OUTPUT:

? (172.16.11.109) at 00:27:0e:13:f7:8c on enp3s0

2) hostname.

OUTPUT:

localhost.localdomain.

3) netstat -r.

OUTPUT:

Kernel IP routing table

Destination	Gateway	Genmask	Flags	MSS
default	gateway	0.0.0.0	UG	0

4) nslookup www.google.com.

OUTPUT:

Server: 172.16.8.1
Address: 172.16.8.1 #53

5) # ping localhost

64 bytes from localhost (::1): icmp_seq=1 ttl=64 time = 0.069 ms

6) ip address show : [Show IP address assigned to an interface]

output :

lo < LOOPBACK, UP, LOWER_UP >

inet 127.0.0.1 /8 scope host lo

inet6 ::1/128 scope host

7) ip address add 192.168.1.254/24 dev enp053

8) ip address del 192.168.1.254/24 dev enp053

9) ip link set eth0 up

10) ip link set eth0 down.

11) ifconfig → replaced by ip command

12) mtr (Matts traceroute) → combines the functionality of the ping and traceroute

mtr google.com

Host	Loss%	Snt	Last	Avg	Best
-gateway	0.0%	182	0.9	1.5	0.3

13) tcpdump → designed for capturing and displaying packets.

a) tcpdump -D

output:
1. enp053
2. any
3. lo.

b) tcpdump -i enp053 : capture all packets on your interface

c) tcpdump -i enp053 -c 10 : Capture traffic to and from host

d) tcpdump -i eth0 -c 10 host 8.8.8.8 : traffic coming from 8.8.8.8

e) tcpdump -i eth0 dst host 8.8.8.8 : traffic going to 8.8.8.8

f) tcpdump -i enp053 net 10.1.0.0/24 : capture traffic to and from a network.

g) tcpdump -i enp053 port 53 : capture only DNS port 53 traffic

h) tcpdump -i enp053 host 8.8.8.8 and port 53 : for a specific host

i) tcpdump -i enp053 -c 10 host www.google.com and port 443 : Captures only "HTTPS" traffic.

j) tcpdump -i enp053 port not 53 and not 25 capture all port except 25 and 53.

14) ping : ~~Root~~ Tool. to verify IP - level connectivi
To another TCP/IP computer by sending
request message via Internet control message
Protocol (ICMP).

ping google.com

Output:

PING google.com (216.58.206.174) 56(84) bytes
of data.
64 bytes from 8of02527-in-f14.1e100.net
(216.58.206.174) : time = 10.7 ms
.
.
.

~~(b) Traceroute connection~~

RESULT:
Thus, the networking commands
has been executed. Successfully.